

SATYAM SINGH

Boston, MA | +1 (857) 398 0665 | satyamsingh.tcet@gmail.com | [LinkedIn](#) | [GitHub](#) | [Website](#)

Generative AI Engineer Experience with researching, developing, and evaluating LLM-powered systems RAG Pipelines, GraphRAG (basic), fine-tuning, prompt engineering, and model evaluation – applied to real world mission-critical workflows. Experienced conveying complex AI concepts to both technical and non-technical audiences.

WORK EXPERIENCE

AI Engineer Co-op | Aosenuma, Boston, MA

April 2025 - September 2025

- Reduced enterprise processing time by 85% across 5k+ users by architecting production multi-agent LLM systems with tool calling and workflow orchestration using **LangChain**, **LangGraph**, and **Anthropic Claude**.
- Prototyped and evaluated GenAI capabilities against real-world workflow needs — benchmarking LLM components using golden datasets, providing actionable recommendations for improving model performance, and conveying findings to non-technical stakeholders through structured evaluation reports
- Engineered RAG and GraphRAG-style pipelines with semantic vector search (ChromaDB, FAISS, pgvector, Neo4j knowledge graphs) and transformer embeddings across 50,000+ domain documents, improving retrieval precision by 45%
- Cut GenAI **API** costs by 30% by designing LLM evaluation frameworks with golden datasets, offline benchmarking, and prompt engineering optimization — maintaining 99.9% uptime across 10,000+ daily requests on **AWS** (Lambda, ECS, SQS)
- Improved API throughput by 40% by delivering 15+ production **REST** APIs (Python **FastAPI** + WebSocket + **Redis**) with async processing, connection pooling, and real-time streaming infrastructure

Software Developer I | LTIMindtree, Mumbai, India

October 2021 - December 2023

- Reduced metadata tracking errors by 80% by engineering an **NLP**-powered document intelligence pipeline with semantic Process ID embedding — adopted as the org-wide production standard across all Paramount systems
- Cut query latency by 25% by designing and optimizing **SQL** Server and PostgreSQL stored procedures and **indexing** strategies across high-throughput pipelines handling millions of daily records
- Reduced MTTR from 45 to 8 minutes (82% improvement) through observability tooling, proactive alerting, and Jenkins **CI/CD** automated test suites maintaining 99.9% uptime for business-critical **enterprise** systems
- Built and shipped production **C#**, **Python**, and **Java** enterprise applications — tax withholding, employee portals, budget workflows — for 500+ **stakeholders** following design patterns, code reviews, and SDLC best practices

Graduate Engineer Trainee | LTIMindtree, Mumbai, India

June 2021 - October 2021

- Completed intensive engineering training program covering Angular, Java, **OOP** design principles, Data Structures & Algorithms, Jenkins **CI/CD**, **Docker containerization**, and cloud fundamentals
- Built training projects applying full-stack development with Angular **frontend** and Java Spring Boot backend, implementing **OOP** patterns, **unit** testing, and version control workflows under senior engineer mentorship

ACADEMIC PROJECTS

GraphRAG Knowledge Intelligence Platform — ESG Document Analysis

April 2025 - June 2025

- Deployed end-to-end RAG system (**LangChain** + **LlamaIndex** + **ChromaDB**) on **AWS Kubernetes** with A/B evaluation pipeline, **human-in-the-loop** feedback, and **DevSecOps** **CI/CD** — achieving 90% test coverage and 30% deployment stability improvement

IBM Watsonx Conversational AI

September 2025 - December 2025

- Built production CPL-driven **agentic** pipeline (**Watsonx.ai**) cutting domain expert resolution from 2 weeks to minutes across 500+ concurrent sessions with 65% latency reduction via async processing and semantic caching

TECHNICAL SKILLS

Languages & Frameworks: Python, TypeScript, JavaScript, Java, C#, FastAPI, Flask, React, Spring Boot, Node.js, LangChain, LangGraph, LlamaIndex, GraphRAG, MLOps, Fine-tuning, Human Language Technology, Model Training

Cloud & DevOps: AWS (Lambda, ECS, SQS, S3, EC2, RDS, Bedrock, SageMaker), Azure, GCP, Docker, Kubernetes, Terraform, GitHub Actions, Jenkins, CI/CD, NVIDIA, GPU acceleration

Data & AI: OpenAI GPT-4, Anthropic Claude, RAG Pipelines, Multi-agent Systems, Prompt Engineering, LLM Evaluation, ChromaDB, FAISS, Pinecone, pgvector, PostgreSQL, MongoDB, Redis, SQL Server

LEADERSHIP & ACHIEVEMENTS

Technical Communication — presented AI system architectures and evaluation findings to cross-functional teams including non-technical stakeholders, translating complex LLM concepts into actionable recommendations

EDUCATION

Northeastern University, Boston, MA

January 2024 - December 2025

Master of Science in Informatics (Cloud Computing)

Mumbai University, Mumbai, India

August 2017 - May 2021

Bachelor of Technology in Electronics & Telecommunications